Entry Without Displacement: How Open-Weight Models Compete in the Market for Inference

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Abstract

Open-weight models are widely expected to constrain the pricing power of proprietary model providers, but the mechanism has not been measured. We assemble a market-wide panel of the models served by a large inference router: posted prices, capability benchmarks, weight-release status and revealed usage for 457 models, extended back to 2023 with archived catalogues. Three facts emerge. Competition works through releases rather than repricing: 54% of models are never repriced at all, and a posted price changes in only 12.6% of model-months, even though changing it costs a provider nothing. When prices do change, volume responds weakly, with an own-price elasticity of -0.53 (s.e. 0.10) identified from 708 revisions with model and date fixed effects. Entry does not displace incumbents: over four weeks, models released inside the window took 28% of tokens while incumbent volume grew 18%, and the confidence interval rules out differential losses above 0.13 log points for the incumbents closest to an entrant in capability space. Removing open-weight models from the choice set in an estimated nested-logit demand system raises the average price paid per token by 156% and costs users about $2.4m per day on this platform, while raising provider concentration. Open weights discipline the price at which new demand is served, not the prices incumbents post.

1 Introduction

Whether open-weight models constrain the market power of proprietary model providers is now a policy question as much as an economic one. Reports commissioned by governments name it as an open evidence gap [Bengio et al., 2026], and the theoretical case cuts both ways: scale economies in training push toward concentration [Korinek and Vipra, 2025], while freely downloadable weights supply a competitive fringe that any hosting provider can serve. Recent empirical work has begun to describe the market. Demirer et al. [2025] document a collapse in the price of a unit of measured intelligence and a proliferation of suppliers; Nagle and Yue [2025] show that a large share of proprietary usage is directed at models that are dominated on price and measured capability by an open alternative, and estimate the savings foregone.

Both of those results are about the allocation of demand across models at a point in time. Neither identifies the channel through which open weights would constrain proprietary pricing, and the two natural channels have opposite implications for policy. If open weights force incumbents to cut posted prices, then the constraint is visible in the price series of established models and its strength can be read off their price responses. If instead open weights capture new demand while incumbents hold their prices, the constraint operates on the terms at which the market grows, and any measurement based on incumbent price responses will find nothing.

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We assemble a panel that can separate the two. Our data come from a large inference router that publishes model-level usage, prices and capability scores. We combine the current cross-section (457 models with observed usage, 175 of them open-weight), a daily usage panel covering all of them for one month, task-level market shares for 29 distinct uses, per-provider hosting terms, and a panel of archived price catalogues and usage leaderboards reaching back to 2023. The panel used for price responses contains 19,803 model-date observations on 634 models.

Three facts follow, and they point the same way. First, the posted price of a given model version behaves like a fixed characteristic of the product: 54% of models are never repriced, and the monthly frequency of a price change is 12.6%, comparable to consumer goods [Nakamura and Steinsson, 2008, Cavallo, 2018] in a setting where the menu cost that explains rigidity there does not exist. Second, when prices do change, volume moves little. Exploiting 708 revisions with model and date fixed effects, we estimate an own-price elasticity of -0.53 (s.e. 0.10), below the value near one reported by Demirer et al. [2025] from cross-model variation and far below what homogeneous-goods reasoning would imply. Third, entry is absorbed by growth rather than by displacement. Over the four weeks of the daily panel the market grew 65%; models released inside the window took 28% of tokens, of which 5 of 10 releases were open-weight, and yet incumbent volume rose 18%. In an incumbent-by-entrant event study we find no differential decline for the incumbents technologically closest to an entrant, with a confidence interval that excludes declines larger than 0.13 log points.

Read together, these facts say that the competitive constraint from open weights does not show up where one would look for it. We therefore quantify it where it does show up. We estimate a nested-logit demand system over the 172 models with prices and capability scores, invert it to recover mean utilities that reproduce observed shares, and delete open-weight models from the choice set. Users would pay 156% more per token, and the surplus loss is worth about $2.4m per day on this platform alone. The loss is not evenly spread: across 29 task-level markets, the implied price increase rises almost one for one with the open-weight share of task spending, from 18% on average to more than twice that in agentic and coding work. Concentration moves the other way from the usual presumption: removing open weights raises the provider Herfindahl index from 897 to 1048.

Our contribution is to identify the margin on which open-weight competition operates and to price it. The measurement of price rigidity and of the within-model demand response is, to our knowledge, new to this market, and it explains why the reallocation gains documented by Nagle and Yue [2025] persist: with inelastic short-run demand, a provider has little to gain from cutting the price of a model already in the field, and users have little reason to move quickly. We also release the collection code, which reconstructs the entire panel from public endpoints.

2 Data

Sources. We use the public API and web endpoints of OpenRouter, a router that serves requests to hundreds of models from many providers through one interface. Four endpoints supply the raw material: a catalogue giving posted input and output prices, context length, release timestamp and the Hugging Face repository for models whose weights are published; a leaderboard giving tokens, requests and tool calls by model; a benchmark table reporting third-party capability scores; and a task table giving each model's share of spending in 29 usage categories over a 30-day window. Each model's own page embeds a daily usage series for the trailing month, which we collect for every model with observed usage. We take per-provider hosting terms from the endpoint listing, and parameter counts from Hugging Face. Historical coverage comes from Internet Archive captures of the catalogue and of the leaderboard; the latter embed the leaderboard payload in the page's streamed data, which we parse directly. Appendix A documents each step.

Openness. We classify a model as open-weight if the catalogue records a public weight repository for it. The classification is model-specific rather than provider-specific, which matters here: several providers, including Google, OpenAI and Alibaba, ship both open-weight and proprietary models, and a provider-level rule would misclassify them.

Capability. Capability enters as a vector of third-party scores covering reasoning, coding and agentic use, supplemented by pairwise-comparison scores from a design-oriented arena. These measures are highly collinear. The first principal component of the three benchmark dimensions explains 98% of their variance, and 91% when the two arena blocks are added, so we summarise capability

Table 1: The estimation sample, measured on the last complete day of the daily panel. Prices are dollars per million tokens for the blended input-output mix. Capability is the first principal component of the benchmark block. Shares are within the estimation sample, which covers 91% of platform tokens.

	Models	Median price	Median capability	Median context (thousands)	Token share	Spend share
Open-weight	86	0.29	30.5	256	0.73	0.32
Proprietary	86	1.50	36.6	400	0.27	0.68
All scored	172	0.59	33.4	262	1.00	1.00

by that first component and report the underlying dimensions in the appendix. Scores are missing for smaller models; we impute a missing dimension by ridge regression on the dimensions a model does have, which recovers held-out values with R^2 above 0.96. The resulting sample of 180 scored models accounts for 91% of tokens.

Prices. Providers post separate input and output prices. Because the input-output mix differs sharply across models, we work with a blended price that weights the two by the observed token mix, and with a fixed-weight index that holds the mix at its sample mean when we need a price series free of feedback from usage. Table 1 summarises the estimation sample.

What the market is. The platform is one segment of the market for inference. It excludes traffic sent directly to a provider's own API and consumer subscription products, and its users are developers who have chosen a router, which selects for price sensitivity and for willingness to switch. Our estimates therefore describe the segment in which open-weight substitution is easiest. If anything this biases us toward finding a strong competitive constraint, which sharpens the interpretation of the weak responses we report below.

3 Three facts

3.1 Capability is cheap, and open weights are cheaper

Figure 1 places every model with usage on the price-capability plane. Two features stand out. At any capability level below the frontier, prices differ by one to two orders of magnitude, so posted price is far from a sufficient statistic for what users buy. Second, the open-weight side of the market is systematically cheaper. Matching each proprietary model to the open-weight models within two points of it on the capability index, the median proprietary model costs 3.3 times the median open alternative, or 1.7 times when the comparison is weighted by volume. Averaged over actual usage, a token from an open-weight model costs \$0.35 per million against \$1.95 for a proprietary one. These gaps are smaller than the ninety percent difference reported by Demirer et al. [2025] and Nagle and Yue [2025] for 2025 data, which is what one would expect if proprietary providers have since released cheaper tiers; the frontier comparison drawn in Figure 1, which uses only the cheapest option available at each capability level, narrows the gap further to 1.4 times.

The right panel tracks the best capability obtainable from each side over time. The open-weight frontier follows the proprietary frontier with a median lag of 105 days. The gap is a lag, not a level: at the very top of the capability distribution proprietary models are unmatched at every date, but the set of tasks for which that matters shrinks as the open frontier advances.

The open-weight share of volume is also far from stable. In the archived leaderboards, which are internally consistent over time though not directly comparable in level with the daily panel, open-weight models held 86% of tokens in late 2023, fell to 29% by mid-2025, and recovered to 48% by mid-2026. The trough coincides with the sample used by Nagle and Yue [2025], whose finding that proprietary models dominate usage describes a market that has since moved.

Openness also changes the supply side. Weighting by volume, an open-weight model is served by 14.9 independent hosting providers against 3.0 for a proprietary model, and the price a user pays for an open model is set by hosts competing to serve the same weights rather than by the organisation that trained it. Open-weight models account for 68% of tokens but only 30% of spending, and the

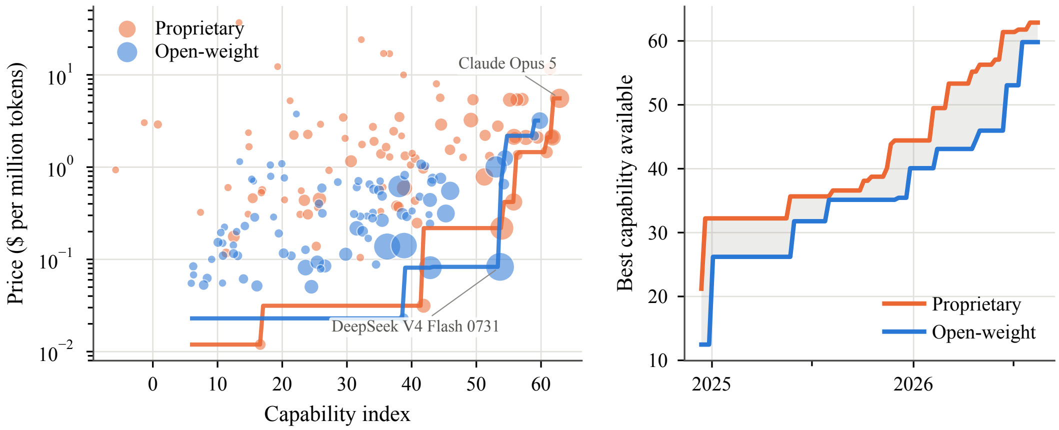

Figure 1: The price of capability. Left: posted blended price against the capability index, with marker area proportional to the square root of token volume; the step functions give the cheapest price available at or above each capability level for each side of the market. Right: the highest capability available from each side by release date, with the shaded region marking the open-weight lag.

difference shows up in concentration: the provider Herfindahl index is 1183 measured on tokens and 2439 measured on spending. By volume the market looks competitive; by revenue it does not.

3.2 Posted prices rarely move

The archived catalogues let us follow the posted price of a model version from its first appearance. Between consecutive captures, a median of three days apart, the price changes 2.9% of the time. Over a model's observed life, 54% of models never change price at all. The left panel of Figure 2 shows the survival of the launch price: 56% of models are still at their launch price after six months and 45% after a year. Conditional on a change, the median revision is a cut of about 5 log points, and 58% of changes are cuts.

Expressed the way the price-setting literature does, 12.6% of model-months contain a price change, an expected duration of about 7.4 months. That is at the low end of the range reported for consumer goods and comparable to online retail [Bils and Klenow, 2004, Nakamura and Steinsson, 2008, Cavallo, 2018], which is itself the surprise: the menu cost that explains rigidity in those settings, the physical and managerial cost of changing a posted price, is close to zero when the price is a field in an API catalogue. The distribution is also more polarised than in retail, where most goods are repriced eventually. Here more than half of models are never repriced at all. What is missing is not the ability to reprice but the incentive, which the next fact supplies. The consequence for measurement is immediate: over the few weeks that a release event study typically spans, an incumbent's posted price is unchanged with probability near 0.9, so a design that looks for the competitive effect of open weights in incumbent price series is looking at a variable that rarely moves.

3.3 Demand for a given model is price-inelastic in the short run

We use the rare revisions to estimate how volume responds. Merging the archived leaderboards with the archived catalogues gives 19,803 model-date observations on 634 models between November 2023 and May 2026, containing 708 observations that follow a price change. We regress log volume on the log fixed-weight price with model and date fixed effects, so identification comes only from within-model revisions, net of anything common to the market on a date:

$$\log y_{jt} = \beta \log p_{jt} + \alpha_j + \gamma_t + \varepsilon_{jt}. \tag{1}$$

Table 2 reports $\hat{\beta} = -0.53$ (s.e. 0.10) for tokens and -0.46 (s.e. 0.09) for requests. Open-weight and proprietary models give indistinguishable estimates. Dropping the model fixed effects, so that cross-model variation identifies β, moves the estimate to -0.24, which is what one would expect if higher-priced models carry unobserved quality.

The right panel of Figure 2 traces the dynamics around 306 revisions, restricted to the captures that report non-overlapping weekly buckets so that the path is not an artefact of a trailing aggregation

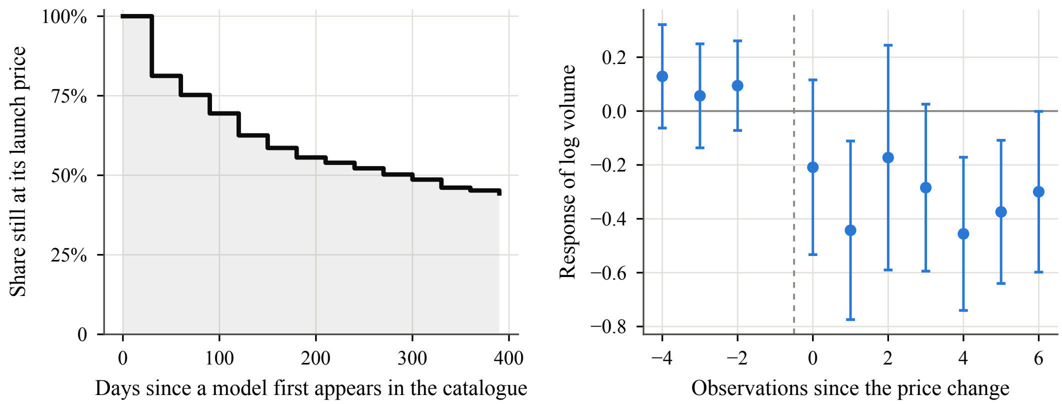

Figure 2: Posted prices rarely move, and volume responds weakly when they do. Left: Kaplan-Meier survival of a model's launch price in the archived catalogues. Right: coefficients from a distributed-lag version of (1) around 306 price revisions, scaled by the size of the revision, with model-by-event and date fixed effects and 95% intervals clustered by model.

Table 2: Own-price elasticity of demand. Each column is a separate regression of (1). Standard errors clustered by model in parentheses.

	Model and date FE		By openness		Date FE only
	Tokens	Requests	Open	Proprietary	Tokens
log price	−0.534	−0.463	−0.582	−0.534	−0.239
	(0.102)	(0.087)	(0.116)	(0.204)	(0.077)
Observations	19,803	19,803	8,873	10,930	19,803
Models	634	634	321	316	634

window. Volume is flat before the change and settles about 0.3 to 0.45 log points lower per log point of price increase within a few weeks. The pre-period coefficients are small and individually insignificant, which is the relevant check: if providers raised prices after volume rose, the estimate would be biased toward zero, and we find no such pattern.

An elasticity near one-half is not consistent with a static Bertrand equilibrium in posted prices, which would require demand to be elastic at the optimum. We read it as a short-run object. Switching a production workload to another model requires re-testing prompts, harnesses and evaluation suites, so the relevant margin adjusts over months rather than days. It also rationalises the rigidity: a provider that would gain little volume from a price cut, and lose little from holding steady, has weak incentive to reprice a model already in the field.

4 Entry without displacement

The daily panel covers every model with usage for the four weeks to 20 August 2026, a window containing 10 releases by providers with priced, scored models, of which 5 are open-weight. The window is unusual in one respect that turns out to be the point: the market grew 65% over it.

The left panel of Figure 3 decomposes daily volume into incumbents and models released during the window. Entrants reached 28% of platform tokens within four weeks, and 69% of that volume went to open-weight entrants. Incumbent volume did not fall. It grew 18%. Entry accounts for 72% of net growth, and the residual is incumbent expansion.

Aggregate volume can hide reallocation, so we test for it directly. For each release k at date T_k we stack all incumbents j that were already serving traffic, and estimate

$$\log y_{jt} = \beta\, \mathbf{1}\{t \geq T_k\} \times \tau_{jk} + \delta_{jk} + \lambda_{kt} + X'_{jt}\theta + \varepsilon_{jkt}, \tag{2}$$

where δ_{jk} is an incumbent-by-entrant effect and λ_{kt} an entrant-by-day effect, so that β compares incumbents exposed to the same release on the same day. The exposure measure $\tau_{jk} = S_{jk}\, r_{jk}$

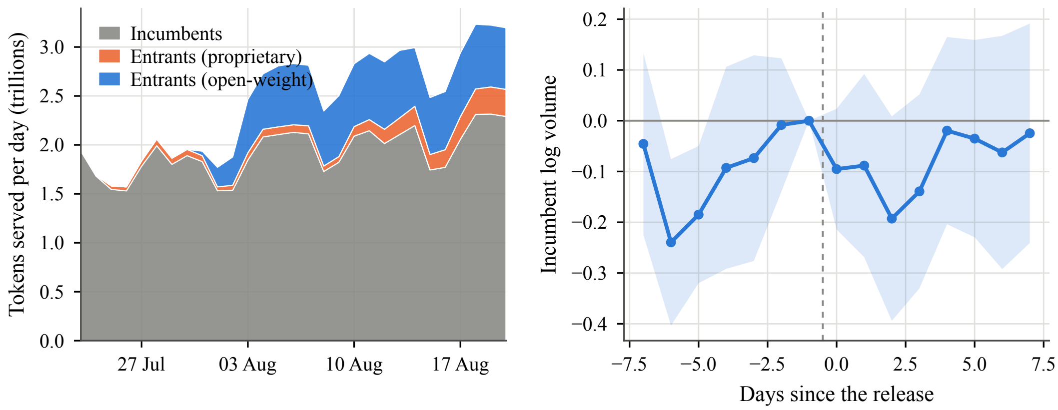

Figure 3: Entry is absorbed by growth. Left: daily platform volume split between models already serving traffic at the start of the window and models released during it. Right: event-time coefficients from (2) on log incumbent volume, per unit of exposure to the entrant, with 95% intervals clustered by incumbent.

combines technological proximity with the entrant's relative value for money: $S_{jk} = \exp(-\kappa d_{jk})$ with d_{jk} the Mahalanobis distance between capability vectors, and $r_{jk} = \log(q_k/p_k) - \log(q_j/p_j)$. Because capability and price both trend, we also control for post-release interactions with the incumbent's own capability and price.

We find no diversion. The estimate of β is 0.01 (s.e. 0.07), and the event-time path in the right panel of Figure 3 is flat on both sides of the release. The confidence interval excludes differential declines larger than 0.13 log points per unit of exposure, and tightens to 0.10 and 0.04 under the Euclidean and cosine distance measures reported in Table 4. A design with ten releases over four weeks cannot detect small effects, and we do not claim that diversion is zero. Nor can this design rule out that exposure is measured with error: if capability similarity is a poor proxy for substitutability, the null is uninformative about diversion even though it is informative about the aggregate pattern. What we can say is that in a market growing at this rate, the displacement channel is small relative to the expansion channel, which is the comparison that matters for how the constraint operates.

A longer horizon. Four weeks is short, and displacement plausibly takes quarters. The archived panel lets us look further out. We take the 19 open-weight releases in it that reached at least one percent of platform tokens, and compare aggregate volume of models already in the market over the eight weekly observations before and after each. Incumbent volume fell in 32% of these episodes; the median change was $+0.25$ log points against a median market change of $+0.52$. The release of DeepSeek R1 is the sharpest case: platform volume rose 0.56 log points over the sixteen weeks around it while incumbent volume was flat at -0.03. Incumbents lose share, because the market grows around them, but for the most part they do not lose volume.

5 What open weights are worth

Because posted prices do not respond and incumbents are not displaced, the value of open-weight competition has to be measured on the choice set rather than on prices. We estimate a nested-logit demand system over the 172 models with prices and capability scores, nesting models by capability tercile so that models of similar capability are closer substitutes than models drawn from different tiers. Writing $g(j)$ for the nest of model j and $\sigma \in [0, 1)$ for the within-nest correlation [Cardell, 1997, Berry, 1994], shares take the form

$$s_j = \frac{\exp\big(\delta_j/(1-\sigma)\big)}{D_{g(j)}} \cdot \frac{D_{g(j)}^{1-\sigma}}{1+\sum_h D_h^{1-\sigma}}, \qquad D_g = \sum_{k\in g} \exp\big(\delta_k/(1-\sigma)\big), \tag{3}$$

with mean utility $\delta_j = \beta_p \log p_j + \alpha' x_j + \xi_j$, where x_j collects the capability index, context length, modality, reasoning support, latency and age. The outside option is the fringe of models we cannot score, which holds 2.4% of tokens.

Two features of (3) make the exercise tractable. Observed shares invert to mean utilities, $\delta_j = \log s_j - \log s_0 - \sigma \log s_{j|g(j)}$ [Berry, 1994], so the system reproduces the data exactly and no part of the counterfactual depends on how well the characteristics fit. And the counterfactual depends only on (β_p, σ): deleting the open-weight models from the choice set changes the denominators in (3), redistributing their volume across the remaining models and the outside option.

Choosing the parameters. We do not attempt to identify β_p and σ from cross-sectional share variation, because the instruments this literature relies on fail in this market. The differentiation instruments of Gandhi and Houde [2019] are functions of the distribution of rival characteristics, which here is a distribution over an essentially one-dimensional capability index, so they are near-collinear with a model's own capability and therefore with its unobserved quality. The obvious cost shifter, parameter count, is available only for open-weight models and predicts unmeasured quality about as strongly as it predicts serving cost: in our data it correlates 0.65 with the capability index, and instrumenting with it moves the price coefficient from -0.79 under ordinary least squares to -0.27 despite a first-stage F above 900, which is the signature of a violated exclusion restriction rather than of a corrected bias. We therefore fix $\beta_p = -1$ and report every result across a range of σ. The diversion test of Section 4 bears on σ from the other direction: finding no differential loss for incumbents in the entrant's own capability tier is what proportional reallocation, σ near zero, predicts. Since the surplus loss is largest at $\sigma = 0$, taking $\sigma = 0.4$ as the headline is again the conservative reading. That value is deliberately conservative: it exceeds in magnitude both the within-model estimate of Section 3.3 and the adjustment-corrected long-run elasticity of about -0.6 implied by (4) below, and it is close to the value reported by Demirer et al. [2025]. Because surplus enters as a log-sum divided by $|\beta_p|$, a larger magnitude yields a smaller dollar figure, so the choice understates rather than inflates what open weights are worth.

Results. Removing open-weight models from the choice set at posted prices, the average price paid per token rises from \$0.70 to \$1.80, an increase of 156%. Consumer surplus, computed as the log-sum expected utility of Small and Rosen [1981], falls 21%, worth about \$2.4m per day at the platform's current volume. The percentage change in surplus does not depend on β_p, which enters the baseline and the counterfactual identically; only the conversion into dollars scales with it. The loss is larger when substitution within a capability tier is weaker, ranging from \$3.9m to \$1.3m per day as σ moves from 0 to 0.7 (Figure 4 and Table 5). It also depends on how large the outside option is taken to be, which is a modelling choice rather than a measurement: raising the outside share from the observed 2.4% to 50% raises the proportional surplus loss, from 21% to 47%, while lowering the dollar figure to \$1.0m per day. The price effect is invariant to both, because it is governed by the composition of the surviving choice set rather than by the strength of substitution. Concentration moves against the presumption that open weights fragment the market: the provider Herfindahl index rises from 897 to 1048 when open models are removed, because open-weight volume is spread across more suppliers than the proprietary volume that replaces it.

These numbers hold prices fixed. That is the right convention given the rigidity documented in Section 3.2, and it makes the exercise a statement about variety rather than about pricing conduct, but it understates the long-run effect. Allowing proprietary providers to reprice would require a pricing model, and the elasticity in Section 3.3 is not consistent with the static Bertrand model that would ordinarily supply one, so we leave that margin unpriced and read our estimates as a lower bound.

The left panel of Figure 4 repeats the exercise separately in each of the 29 task-level markets, where shares of spending are reported directly. The implied price increase rises almost one for one with the open-weight share of task spending, with a slope of 0.98. Averaged across tasks and weighted by spending, prices rise 18% and surplus falls 13%, with the largest effects in agentic and coding work, where open-weight models hold about a third of spending, and the smallest in data extraction and transformation, where they hold under a sixth. The task-level numbers are smaller than the aggregate because task shares are measured on spending, which weights the expensive proprietary models that open weights do not displace.

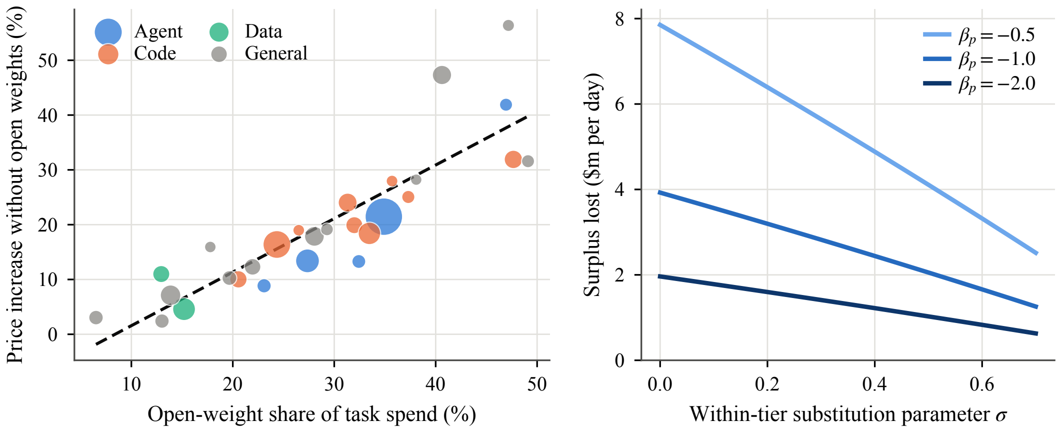

Figure 4: The value of the open-weight choice set. Left: for each of the 29 task markets, the price increase implied by removing open-weight models against their share of task spending; marker area is proportional to the task's share of platform spending. Right: aggregate surplus loss per day across values of the within-tier substitution parameter and the price coefficient.

6 Discussion

The three facts fit together. Short-run demand for a given model is inelastic, so providers gain little by cutting the price of a model in the field and do not; because posted prices do not move, competition is expressed in the terms on which new models enter; and because the market is growing quickly, entrants can take a large share of volume without taking it from anyone. In that environment, the standard empirical signature of competitive discipline, a price response by incumbents, is absent even though the constraint is economically large. Our counterfactual puts the value of the open-weight choice set at roughly one fifth of consumer surplus on this platform.

A simple account. The three fit together through the share of demand that is actually in play. Let M_t be total demand, growing at rate g per period, and let a fraction ϕ of existing demand be revisited each period, whether because a workload is rebuilt or because a team re-evaluates its choice. Contestable demand is $(g + \phi)M_t$, and a model's volume evolves as

$$y_{j,t+1} = (1 - \phi)\, y_{jt} + \omega_j\, (g + \phi)\, M_t, \tag{4}$$

where ω_j is its share of contestable demand. Three implications follow. An incumbent loses volume only when $\omega_j(g+\phi)M_t < \phi\, y_{jt}$, that is, when its share of contestable demand falls below $\phi/(g+\phi)$ times its share of the stock; with $g \approx 0.13$ per week in our window, an incumbent with an average share of new demand keeps growing. An entrant reaches a share of about $\omega_k(g + \phi)/(1 + g)$ after one period, so entrants can take a large share of volume quickly without anyone shrinking. And a permanent price change reaches only the contestable margin, so the response measured τ periods out is attenuated by roughly $1 - (1 - \phi - g)^\tau$ relative to its long-run value.

That last point bounds how much of the inelasticity is an artefact of slow adjustment. Taking $\phi + g \approx 0.23$ per week and the six weekly observations over which Figure 2 shows the response settling, the attenuation factor is about 0.8, so the long-run elasticity implied by our estimate is near -0.6. Demand for a given model is inelastic even after adjustment. The static first-order condition of a provider maximising current profit is therefore not satisfied at the posted price, which leaves two readings: providers price against a margin we do not observe, such as capacity, hardware amortisation or the option value of usage; or the posted price is not chosen to maximise current profit at all. The rigidity in Section 3.2 favours the second.

This also reconciles two readings of the existing evidence. Nagle and Yue [2025] find that a great deal of proprietary usage sits on models dominated by an open alternative, and infer large foregone savings. That pattern is a puzzle if demand adjusts quickly; it is what one should expect if the short-run elasticity is about one-half and switching a workload is costly. The gains they identify are real, but they accrue as workloads turn over rather than as incumbents reprice.

Limitations. Our market is one router, and one that selects for price-sensitive developers, so the elasticities and the counterfactual describe the segment where substitution toward open weights is easiest. The capability index is built from third-party benchmarks that compress into a single dimension, and models that look identical on it are plainly not interchangeable in practice, which is precisely why the discrete-choice approach is needed and also why the index should not be read as a complete description of quality. We do not identify β_p and σ from the data and instead report sensitivity across a range; a design with exogenous variation in the choice set, such as a large release observed over a longer window, would pin them down. The diversion test has power against moderate effects only. Finally, the counterfactual holds prices fixed, and is therefore silent about how the proprietary frontier would be priced in a world without open weights.

Implications. For policy, the result reframes what to measure. Debates about the competitive effect of open weights have proceeded on the presumption that the effect, if real, would show up in the prices proprietary providers charge. In a market where posted prices are set once and left alone, that presumption is wrong, and an assessment based on incumbent price responses will conclude that open weights do not matter. The relevant margin is the price at which the next unit of demand is served, and on that margin the constraint is substantial.

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A Data construction

Endpoints. The catalogue endpoint returns, for every listed model, the posted input and output price per token, context length, input and output modalities, a creation timestamp, and the Hugging Face repository identifier when the weights are public. The leaderboard endpoint returns tokens, requests, tool calls and media counts by model and serving variant. A benchmark endpoint returns third-party capability scores and usage-weighted effective prices. A task endpoint returns, for each of 29 usage categories, the category's share of platform spending and the share held by each of its ten largest models, in both spending and token terms. An endpoint listing gives each hosting provider's price, quantisation and uptime for a given model. All are public and unauthenticated.

Daily usage. Each model's public page embeds its own trailing-month daily usage series in the page's streamed render payload. We request one page per model with observed usage and parse the payload, which yields 457 models over 30 days. The same payload carries per-endpoint latency and throughput percentiles, which we aggregate to the model level by request count.

Historical coverage. Internet Archive captures supply history. Captures of the catalogue endpoint give a price and characteristic panel from July 2023. Captures of the leaderboard page embed the leaderboard payload in the same streamed format, so we recover model-level usage without relying on the rendered page. Two vintages are present and we treat them separately: captures before late 2024 embed a series of non-overlapping weekly buckets, and later captures embed a single trailing-window snapshot. Shares are comparable across vintages; levels are not, so all specifications include date fixed effects, and Table 3 reports the two vintages separately. We fetch captures at a fixed spacing with adaptive backoff, which is the binding cost of the collection: the archive requests take a few hours, and every other step runs in minutes on a laptop.

Merging. Usage and price observations are matched by model identifier and nearest date within ten days. Openness is taken from the current catalogue where the model is still listed, because the weight-repository field is absent from the earliest captures. Prices are dollars per million tokens throughout. The fixed-weight price index used in (1) weights posted input and output prices by the sample-mean token mix, so that it carries no feedback from a model's own usage composition; Table 3 shows the estimate is not sensitive to that choice.

B Capability imputation

Benchmark scores are missing for smaller models. For each missing dimension we fit a ridge regression on the dimensions the model does have, using models scored on both, and predict the missing value. Masking half of the fully scored models and predicting the held-out values gives R^2 of 0.99, 0.97 and 0.96 for the reasoning, coding and agentic dimensions, with mean absolute errors of 1.7, 5.0 and 3.1 points against standard deviations of 16.4, 22.6 and 18.5. Imputation raises coverage from 109 models and 77% of tokens to 180 models and 91% of tokens. All results in the main text hold on the directly scored subsample.

C Robustness

Table 3 reports the elasticity across price measures, outcome measures, capture vintages, openness and sample period. The estimate stays between -0.38 and -0.63. The realised blended price, which mixes posted prices with the model's own input-output composition, gives the largest magnitude, as expected if a shift toward prompt-heavy workloads lowers the average price per token and raises volume at the same time; the fixed-weight index removes that channel and is our preferred measure.

The estimate is not driven by any one model: dropping each of the 268 models whose price ever moves, one at a time, leaves it between -0.58 and -0.51. Restricting to models holding at least 0.01% of tokens gives -0.378 (s.e. 0.127).

Two further checks bear on the elasticity. A price cut may accompany the launch of a replacement by the same provider, which would depress the old model's volume for reasons other than price. Adding an indicator for observations within 28 days of a same-provider release, which covers 38% of the panel, leaves the estimate at -0.534; dropping those observations gives -0.499 (s.e. 0.123).

Table 4 reports the diversion test under three definitions of technological distance. None of the six estimates is distinguishable from zero. The Euclidean and cosine measures give tighter standard errors than the Mahalanobis measure, and bound the differential decline for the most exposed incumbents at 0.13 and 0.09 log points respectively.

Table 3: Own-price elasticity across specifications. Each row is a separate estimate of (1) with model and date fixed effects; standard errors clustered by model.

Specification	Estimate	Std. error	Observations	Models
Fixed-weight price (baseline)	-0.534	0.102	19,803	634
Input price	-0.526	0.096	19,803	634
Output price	-0.485	0.109	19,803	634
Realised blended price	-0.658	0.095	19,803	634
Requests rather than tokens	-0.463	0.087	19,803	634
Weekly captures only	-0.472	0.123	15,133	432
Trailing captures only	-0.476	0.125	4,670	428
Open-weight models	-0.582	0.116	8,873	321
Proprietary models	-0.534	0.204	10,930	316
2025 onwards	-0.527	0.100	7,906	542

Table 4: Diversion at entry under alternative distance measures. Each row estimates (2) with incumbent-by-entrant and entrant-by-day fixed effects and post-release controls for the incumbent's own capability and price.

Distance	Exposure	Estimate	Std. error	Observations
Mahalanobis	similarity $\times$ relative value	0.010	0.070	11,820
Mahalanobis	similarity	0.019	0.120	11,820
Euclidean	similarity $\times$ relative value	-0.006	0.047	11,820
Euclidean	similarity	0.008	0.065	11,820
Cosine	similarity $\times$ relative value	0.022	0.031	11,820
Cosine	similarity	0.083	0.086	11,820

D Counterfactual sensitivity

Table 5 reports the counterfactual across the within-tier substitution parameter, and `robustness.json` reports it across the size of the outside option as well. The price effect is nearly invariant to σ, because it is driven by the composition of the surviving choice set rather than by the strength of substitution. The surplus loss falls as σ rises, because closer substitutes within a capability tier make the proprietary models better replacements for the open-weight models that leave.

Table 5: The open-weight counterfactual across substitution parameters, at $\beta_p = -1$. Prices are dollars per million tokens; the surplus loss is evaluated at the platform's volume on the reference day.

σ	Price paid ($/M)	Price without open ($/M)	Surplus change (%)	Surplus loss ($m per day)	HHI (without open)
0.0	0.70	1.82	-33.8	3.9	1109
0.2	0.70	1.81	-27.5	3.2	1082
0.4	0.70	1.80	-21.0	2.4	1048
0.6	0.70	1.79	-14.3	1.7	1006
0.7	0.70	1.78	-10.8	1.3	982

E How good and how cheap an open model has to be

The counterfactual removes the open-weight models that exist. The opposite exercise asks what a new one would have to look like to matter. We insert a single hypothetical open-weight entrant with capability $(1-\rho)$ times the frontier and price p, assigning it the mean utility implied by the fitted gradient of δ_j in capability and log price, and recompute shares.

Figure 5 traces the effect on proprietary volume. The iso-effect lines are steep, which is the substantive point: price moves the outcome far more than the last few points of capability. An entrant that matches the frontier exactly but is priced at \$0.50 per million tokens takes 6.6% of platform tokens and cuts proprietary volume by 7%. An entrant a tenth below the frontier but priced at \$0.05 takes 17% and cuts proprietary volume by 18%. At the frontier's own price, an open model that matches it takes essentially nothing. Closing the last of the capability gap is worth much less to the proprietary side than the pricing that open-weight release makes possible, which is consistent with the pattern in Figure 1: the open frontier lags by about a quarter, but sells the capability it does reach at a fraction of the price.

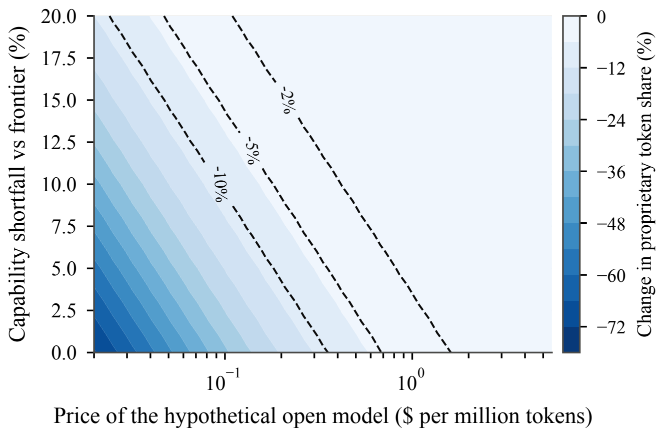

Figure 5: The entrant a proprietary provider should fear. Change in proprietary token share when one hypothetical open-weight model is added to the choice set, by its price and its capability shortfall against the frontier, at $\beta_p = -1$ and $\sigma = 0.4$. Contours mark 2, 5 and 10 percent losses.

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